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Mission Design of a Tethered Robot Satellite "STARS" for Orbital Experiment

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***Abstract*— The Space Tethered Autonomous Robotic Satellite (STARS) was developed in Kagawa University and launched by the H-IIA rocket by the Japan Aerospace Exploration Agency (JAXA) on 23 January, 2009. The primary objective of STARS is technical verification of a tethered space robot, which is connected to a mother spacecraft through a tether, proposed in 1995. Main characteristics of STARS are: (i) it is mother-daughter satellite; (ii) it becomes a tethered system on orbit; (iii) the daughter satellite is a tethered space robot. STARS consists of two satellites connected by a tether. One is a mother satellite (a mother spacecraft), which deploys and retrieves a tether connected to the other satellite. The other is a daughter satellite (a tethered space robot). The mother satellite has a tether deployment system. During tether deployment, the attitude of the daughter satellite is controlled by its own arm link motion, and a camera mounted on the daughter satellite takes a photograph of the mother satellite. It is important to design launch lock for STARS, since it has many actuation parts and it is two satellites system.**

I. Introduction

THE Kagawa Satellite Development Project started since January 2005, in order to develop the Space Tethered Autonomous Robotic Satellite (STARS). STARS was launched on 23 January, 2009 by the H-IIA rocket by the Japan Aerospace Exploration Agency (JAXA) with the main satellite "GOSAT," and piggy bag satellites "SDS-1," "SOHLA-1," "SPRITE-SAT," "SORUN-SAT," "PRISM," and "KKS-1" (see figure 1). Tokyo University developing "PRISM," Tokyo Metoropolitan College of Industrial Technology developing "KKS-1," and Kagawa University are belonging to the University Space Engineering Consortium (UNISEC), organized by Japanese university and college members in the field of space engineering [1]. Up to now, 7 satellites were launched by UNISEC members: "XI-IV" and "XI-V" by Tokyo University, "Cute-1," "Cute1.7+APD," and "Cute1.7+APD II" by Tokyo Institute of Technology, "HIT-Sat" by Hokkaido Institute of Technology, and "SEEDS" by Nihon University.

The primary objective of STARS is technical verification of the capability of a tethered space robot. The tethered space robot is a new type of space robot system proposed in previous work [2], [3]. It differs significantly from the

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Fig. 1. Flight model of STARS

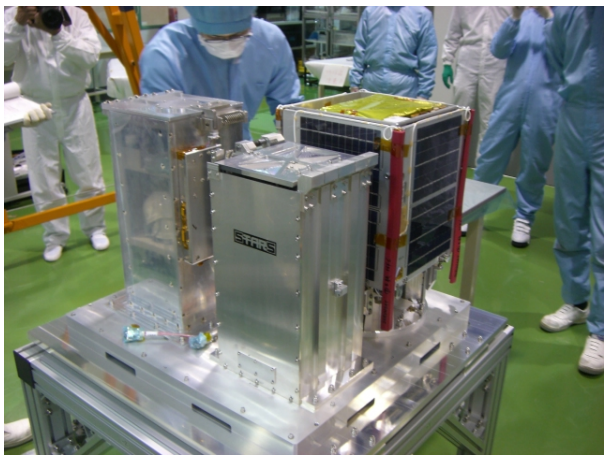


Fig. 2. Nano-satellite group mounted on the H-IIA #15 rocket.

Tethered Satellite System (TSS) studied so far [4], [5] in three aspects. First, we assume that the tether is to be extended at a relatively short distance. Second, we do not envision either gravity force or centrifugal force to be an influence on the tether extension. Instead, we employ a tether extension strategy assisted by the initial translation momentum of the subsystem. Third, we envision the tethered subsatellite to be a multi-body system. The major advantage of the multi-body nature of the subsatellite is that its attitude can be controlled under tether tension by its own link motion. This can be done by employing methods developed in free-flying space robot studies [6], [7].

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Main characteristics of STARS are: (i) it is two satellites system, mother and daughter; (ii) it becomes a 5m tethered system on orbit at the maximum; (iii) the daughter satellite is a tethered space robot, whose attitude can be controlled by its own arm link motion. The mother satellite is a tether deployment system, and it deploys the daughter satellite and retrieves it. The daughter satellite is a tethered space robot, and it has one arm link attached to the end of the tether. Attitude control by arm link motion using tether tension is possible, and so the orientation of the camera mounted on the daughter satellite can be controlled in order to take a photograph of the mother satellite. For cooperation control of tether deployment, attitude control, and camera operation, inter-satellite communication is needed for performing the mission.

Figure 1 shows the flight model of STARS. Here the mother (below) and the daughter (upper) satellites are docking. STARS is inserted in the separation box on the rocket, as shown in Figure 2. This paper describes mission design of STARS especially focusing on tether deployment and attitude control by TSR mechanism. Section II summarizes specification and mission of STARS. Section III explains the mother satellite, which has tether deployment system. Section IV explains the daughter satellite of STARS, which has the arm link mechanism for TSR motion. Section IV explains interface with the H-IIA rocket, and also separation sequence from the rocket.

II. Main Mission for Tethered Space Robot

Figure 3 shows image of STARS on orbit. The specifications of STARS are:

Mother satellite
Mass: 4.2 kg
Length: 160 x 160 x 253 mm
(without solar paddles and corn)

Daughter satellite
Mass: 3.8 kg
Length: 160 x 160 x 158 mm
(without solar paddles and arm link)

Each satellite has 4 subsystems: electrical power subsystem (EPS), communication subsystem (COMM), control & data handling subsystem (C&DH), and camera subsystem (CAMERA). They have gyro, magnetic, acceleration, thermo sensors, and voltage and current sensors. Therefore, the mother and the daughter satellites are operated independently, and they communicate each other through Bluetooth (inter-satellite communication) for tether deployment mission. They have magnetic torquers for attitude control, and also have motors with encoders for tether deployment and robotic motion. Also, each satellite has two solar paddles. Antennas for transmitting telemetry and receiving command are attached on the edge of the paddles, respectively.

As shown in Figure 4, tether deployment mission begins with a command from the ground station (Phase A). First, electrical power is supplied to the control devices on the both satellites. Since tether deployment and robot motion are



Fig. 3. Computer graphics image of STARS

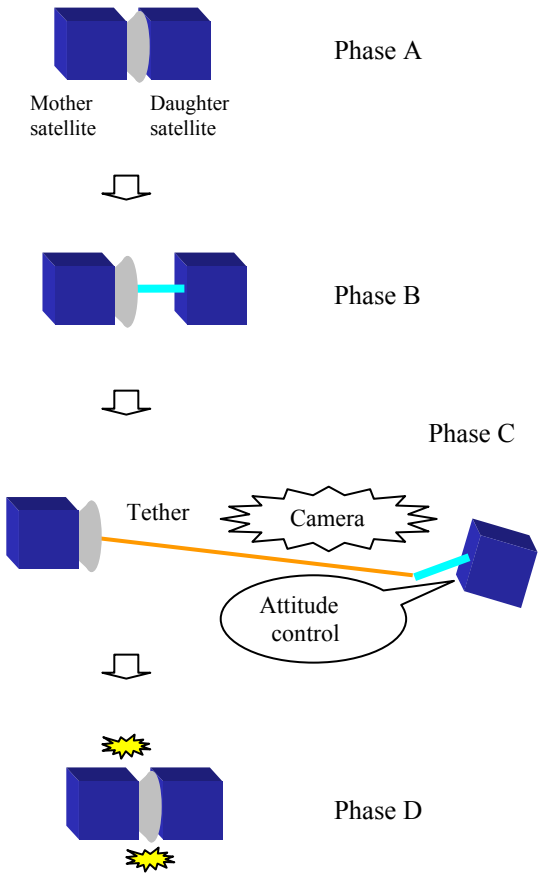


Fig. 4. Mission sequence.

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controlled by motor actuation and need high electrical power, electrical power is supplied only during the mission. Next, the mother satellite releases the daughter satellite, and the daughter satellite begins to control its attitude by its own arm link motion (Phase B). The daughter satellite, whose attitude is controlled during deployment, takes a photograph of the mother satellite when the tether length becomes long enough (Phase C). Finally, the daughter satellite docks with the mother satellite (Phase D). After the mission is terminated, mission data and picture data are transmitted to the ground station.

III. Tether Deployment System

A. Mechanical system

Figure 5 shows tether deployment system equipped on the mother satellite. The 4 top points of the main frame contact with the frame of the daughter satellite. The shape of the docking cone fits the arm link of the daughter satellite: the upper edge contacts with the top cover and the below peak

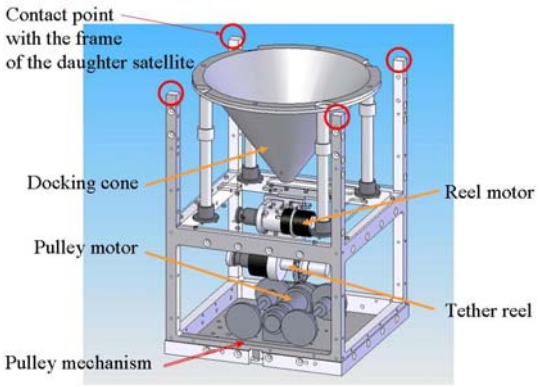


Fig. 5. Tether deployment mechanism on the mother satellite.

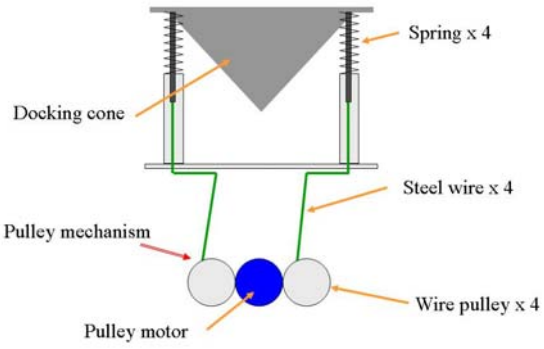


Fig. 6. Docking cone wired to pulley mechanism.

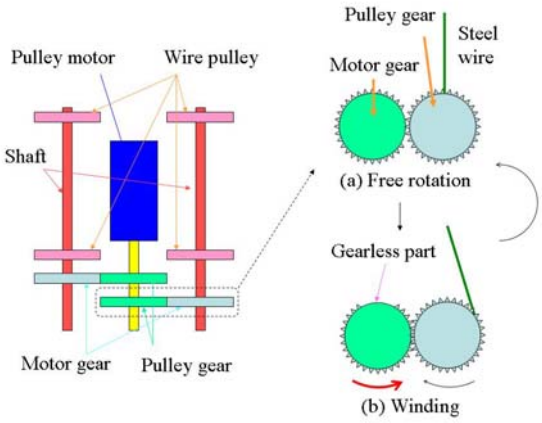


Fig. 7. Motor and pulley gear interface.

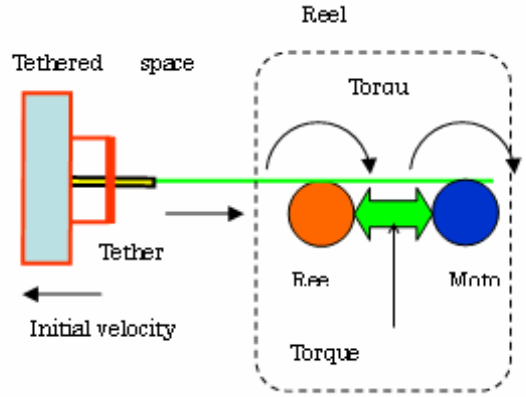


Fig. 8. Tether reel system.

contact with the end of the arm link, respectively. As shown in figure 6, the docking cone is supported by 4 springs, which are compressed by 4 steel wires wound up by 4 wire pulleys, respectively. As shown in figure 7, 2 wire pulleys connected to one pulley gear through one shaft. The wire pulleys are rotated by one pulley motor through motor and pulley gears. The motor gear has gearless part. Under condition (a), wire pulleys can rotate freely. As the motor gear rotates in counterclockwise direction, the pulley gear rotate in clockwise direction and steel wires are wound up. When the gearless part makes one rotation, wire pulleys can rotate freely. Therefore springs are extended and the docking corn is pushed up, then the daughter satellite deploys.

Tether tension is controlled to be constant by tether reel system. As shown in figure 8, the tether reel system consists of tether reel and reel motor connected through torque transmission device, which transmits constant torque from the reel motor to the tether reel, then tether is controlled at constant tension.

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B. *Parameters and sequence*

Parameters of the tether deployment system are follows.

Spring constant:	0.165N/mm
Compression length of the spring:	71.2mm
Tether tension:	27gf

Then, mission will be performed as following condition from dynamics analysis.

Initial velocities of the mother satellite:	0.61m/s
Initial velocities of the daughter satellite:	0.67m/s
Maximum tether length:	6.13m
Mission time:	19.2s

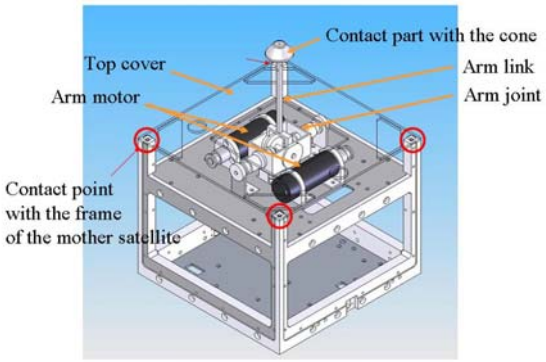


Fig. 9. Arm joint mechanism on the daughter satellite.

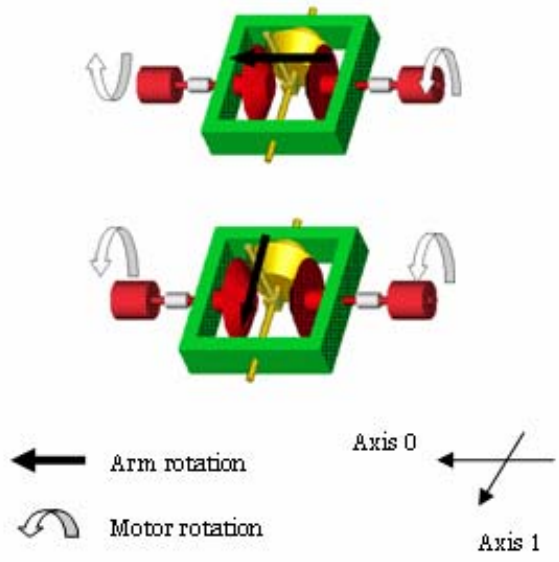


Fig. 10. Differential joint mechanism.

IV. Arm Joint System

A. *Arm Joint Mechanism*

Figure 9 shows arm joint mechanism equipped on the daughter satellite. Tether is attached to the end of the arm link. Two arm motors actuates the arm link by the arm joint mechanism. The 4 top points of the main frame contact with the frame of the mother satellite. The top cover covers the arm joint mechanism and contact with the cone edge of the mother satellite. For the purpose of attitude control of the vertical axes of the tether extension line, the differential gear mechanism is employed as shown in Figure 10. The rotation axes of the two motors are the same, and they are denoted by axis 0 and axis 1, which is vertical to the tether extension line. The two motors are fixed on the satellite body. Both motors rotate as shown in the upper half of figure 10: the arm link rotates around the 1 axis, and when the two motors rotate, as in the lower half, the arm links rotate around the 0 axis.

Gear ratio of the motor and arm is 1:2. Therefore relation of arm angle and motor angle is described by following equations:

$$\phi_0 = (\psi_0 - \psi_1)/2 \quad (1)$$

$$\phi_1 = (\psi_0 + \psi_1)/2 \quad (2)$$

where ϕ_0 and ϕ_1 denote arm angles in the 0 axis and the 1 axis, respectively. ψ_0 and ψ_1 denote motor angles the 0 axis and the 1 axis, respectively.

where θ and ϕ denote base attitude angle and arm angle with respect to the base, respectively.

B. *Arm motion control*

Here explains the attitude control of the daughter satellite by its own arm link motion. First, let us consider a simple planer model for a tethered space robot consisting of one base and one link arm, as shown in Figure 11. The base attitude is

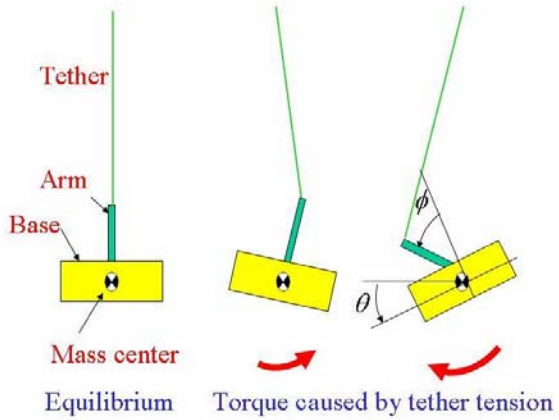


Fig. 11. Attitude Control of tethered space robot.

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in equilibrium when the mass center of the robot is located on the tether extension line. Tether tension torque acts on the robot when the mass center deviates from the tether extension line. Because the tether is attached to the end of the arm link, the tether tension torque can be controlled by an arm link operation. It should be noted that attitude control is possible around the vertical axes with respect to the tether extension line, and the arm link needs two degrees of freedom.

Here, let's consider the following simple PD control for attitude of the base:

$$\phi = k_{p\theta}(\theta - \theta_c) + k_{d\theta}(d\theta / dt), \qquad (3)$$

where θ and ϕ denote base attitude angle and arm angle with respect to the base, respectively. θ_c denotes the desired attitude, and $k_{p\theta}$ and $k_{d\theta}$ denote control gains, respectively. Here, θ_c =0, and $d\theta/dt$ and θ are obtained from measurement by gyro sensors. By employing equation (3) around the 0 and 1 axes, attitude of the base around tether extension line can be controlled. The first term $k_{p\theta}(\theta - \theta_c)$ cannot damp rotational vibration of the base around equilibrium, though it can cause large tether tension torque which makes vibration frequency high. In order to suppress rotational vibration, the second term $k_{d\theta}\, d\theta/dt$ is necessary.

V. Rocket Interface

A. Launch lock mechanism

Launch lock mechanism is necessary for vibration and impact environment during rocket launch. Figure 12 shows the launch lock mechanism for the mother and daughter satellites under docking condition. Horizontal motion is restricted by restriction of main frames with their guides, and paddles are locked by their hinges and lock mechanism. And the mother and daughter satellites under the docking condition are inserted in the separation box fixed on the rocket. By closing the upper cover of the separation box,

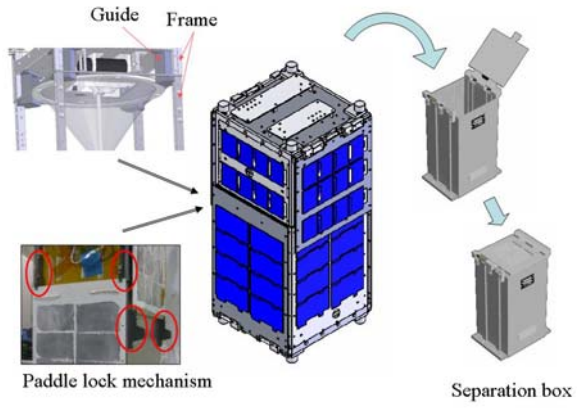


Fig. 11. Launch lock of mother and daughter satellite.

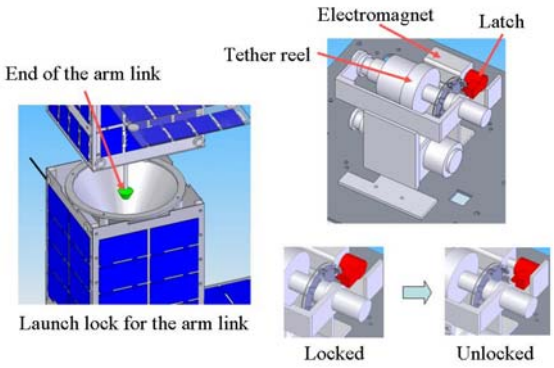


Fig. 12. Launch lock of the arm link and the tether reel.

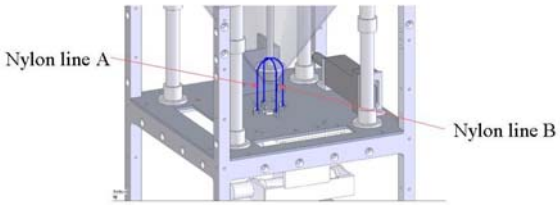


Fig. 13. Nylon lines for locking.

vertical motion of the satellite is also restricted.

The arm link and its joint, and the tether reel are locked as shown in figure 13. Because the shape of the end of the arm link is to be fixed along the docking corn, the arm link and its joint are locked under docking condition of the mother and daughter satellites. The tether reel is locked by a latch controlled by electromagnetic. The latch for the tether reel is also used for nominal orbital operation of docking condition of the mother and the daughter satellites.

The launch lock mechanism was verified by the vibration and impact tests required JAXA.

B. Separation sequence from the rocket

In order to separate from the rocket under docking condition of the mother and the daughter satellites, and to perform to deploy antennas automatically after the separation, two nylon lines are used for locking the mother and the daughter satellite as shown in figure 14. Nylon line A is longer than nylon line B (about 40mm), and they are cut by heater (Nickrome wire).

After the complete separation from the rocket, nylon line B is cut, and the mother and the daughter satellites are detached due to the springs of the corn, and then the paddles are deployed as shown in figure 15, and CW transmission starts. Here, distance between the mother and the daughter satellite is kept at 20mm by nylon line A.

Nylon line A is cut (by heater) by the command from the

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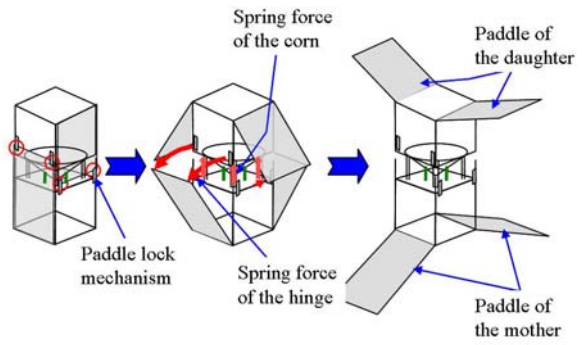


Fig. 14. Unlock sequence of the mother and daughter satellite.

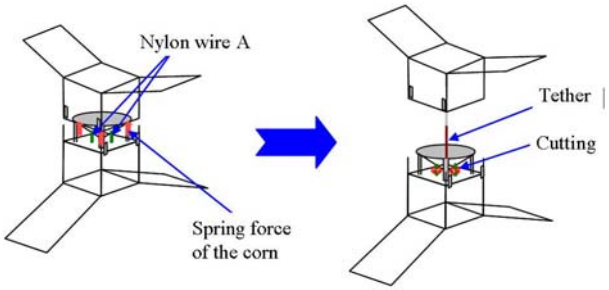


Fig. 15. Unlock sequence of tether deployment.

ground station. Since the springs of the corn are completely extended by cutting nylon line A, the tether is deployed under the following condition as shown in figure 16.

Compression length of the spring:	50.2mm
Initial velocities of the mother satellite:	0.43m/s
Initial velocities of the daughter satellite:	0.48m/s
Maximum tether length:	3.13m
Mission time:	13.7s

This is the first orbital experiment of STARS.

VI. Conclusion

This paper has been described mission design of STARS launched by the H-IIA rocket in the beginning of 2009. Main characteristics of STARS are: (i) it is two satellites system, mother and daughter; (ii) it becomes a 5m tethered system on orbit at the maximum; (iii) the daughter satellite is a tethered space robot, whose attitude can be controlled by its own arm link motion. Structure design and parameters of the mother satellite and the daughter satellite have been described, respectively.

Figure 16 shows the photograph of the rocket with piggy bag satellites taken by GOSAT at its separation from the rocket. It is noted that STARS and other satellites were launched without structural destruction. Also, it was confirmed from the first communication of STARS from the ground station that unlock sequence as shown in figure 14 was successfully performed. And then, it was verified by the first operation that each launch lock mechanism mechanism worked well.

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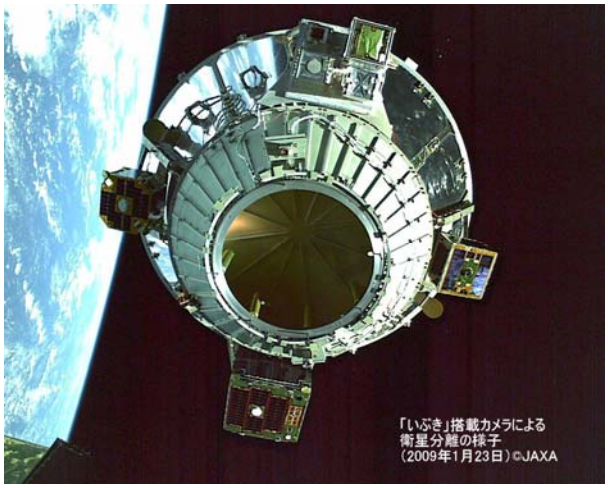


Fig. 16. Photograph taken from GOSAT its separation (©JAXA)

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